Program Demonstrating Polymorphism

```
#include <iostream>
using namespace std;
// ----------------- Compile-Time Polymorphism (Function Overloading) -----------------
class Calculator
{
public:
// Function to add two integers
int add(int a, int b)
{
return a + b;
}
// Function to add three integers (overloading)
int add(int a, int b, int c)
{
return a + b + c;
}
// Function to add two floating-point numbers (overloading)
float add(float a, float b)
{
return a + b;
}
};
// ----------------- Run-Time Polymorphism (Function Overriding) -----------------
class Shape
{
public:
// Virtual function for area
virtual void area()
{
cout << "Area of generic shape" << endl;
}
};
```

```
class Rectangle : public Shape
{
private:
float length, width;
public:
Rectangle(float l, float w)
{
length = l;
width = w;
}
// Overriding the area function
void area() override
{
cout << "Area of rectangle = " << length * width << endl;
}
};
class Circle : public Shape
{
private:
float radius;
public:
Circle(float r)
{
radius = r;
}
// Overriding the area function
void area() override
{
cout << "Area of circle = " << 3.1416 * radius * radius << endl;
}
};
// ----------------- Main Function -----------------
int main()
```

```
{
// Compile-time polymorphism demonstration
Calculator calc;
cout << "--- Compile-Time Polymorphism (Function Overloading) ---\n";
cout << "Sum of 2 integers: " << calc.add(10, 20) << endl;
cout << "Sum of 3 integers: " << calc.add(5, 10, 15) << endl;
cout << "Sum of 2 floats: " << calc.add(2.5f, 3.5f) << endl;
// Run-time polymorphism demonstration
Shape *s1, *s2;
Rectangle rect(10, 5);
Circle circ(7);
s1 = &rect; // Base class pointer pointing to derived class object
s2 = &circ;
cout << "\n--- Run-Time Polymorphism (Function Overriding) ---\n";
s1->area(); // Calls Rectangle's area()
s2->area(); // Calls Circle's area()
return 0;
}
```

OUTPUT

```
--- Compile-Time Polymorphism (Function Overloading) ---
Sum of 2 integers: 30
Sum of 3 integers: 30
Sum of 2 floats: 6
--- Run-Time Polymorphism (Function Overriding) ---
Area of rectangle = 50
Area of circle = 153.938
```